DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #05A: NON-WEAPON EQUIPMENT & PROCEDURAL SYSTEM (PART 1 — BODY ARMOR & HEADGEAR ARSENAL: HEAVY, LIGHT, ROBE, HEADGEAR, ADAPTIVE ARMOR & 3 LEGENDARY ARMORS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap for Series `05A–05G` (Complete Non-Weapon, Procedural & Total Equipment Audit)**:
  - **`05A` (This Document)**: Group 1 — **Body Armor & Protective Attire (`HEAVY`, `LIGHT`, `ROBE`, `HEADGEAR`, Adaptive Armor `E02`, `CELESTIAL_ARMOR_ARTISAN`, & 3 Legendary Armors `LEG_ARM_*`)**
  - `05B`: Group 2 — Personal Jewelry & Wearables (`RING`, `NECKLACE`/`PENDANT`, `BRACELET`, `CHARM`/`ORNAMENT`, `SHOES`, Storage Bags, & 4 Legendary Accessories `LEG_ACC_*`)
  - `05C`: Group 3 — Wearable Talismans, Portable Formation Cores & Profession/Exploration Tools (`ITEM190_TAL_*`, `ITEM190_CORE_*`, `ITEM190_TOOL_*`, `ITEM_V234_TOOL_*`, Cauldrons & Collars)
  - `05D`: Group 4 — Multi-Piece Set & Synergy Architecture (`62` Resonance Sets, Paired Items `E03`, Cursed/Soul-Bound `E04`, Catalysts `1–3★` & Primordial Trinity)
  - `05E`: Part 5 — Holistic Overview of Non-Weapon Equipment (Single `ACCESSORY` Slot Congestion, `charges` vs `durability` Split, Work-Tool vs Combat-Gear AI Switching)
  - `05F`: Part 6 — Penultimate Deep Audit: Procedural Equipment Engine (`rollProceduralGear`, `ITEM_PROC_*`/`REC_PROC_*` Templates, `slotOptions`, `materialMods`, 12 Affixes, Repair & Salvage)
  - `05G`: Part 7 — Grand Finale: Complete End-to-End Master Audit of the Entire Equipment Ecosystem (`898` Items Across All Systems)

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Canonical Type Normalization**: Gameplay (`martial.js`, `stats.js`, `combat.js`) and Presentation (`ui.js`, `render.js`) must agree 100% on canonical equipment types (`HEAVY`, `LIGHT`, `ROBE`).
3. **100% Emergent NPC Autonomy**: Frontline tanks must favor and maintain `HEAVY` plate, agile assassins/archers must favor `LIGHT` mail, Qi casters/scholars must favor `ROBE` vestments, and armor artisans must forge replacements as frontline gear wears down.

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1. `[ARM-TAG-01]` Critical Catalog-vs-Engine Tag Desync: `'HEAVY_ARMOR'`, `'LIGHT_ARMOR'` & `'HEADGEAR'` Break Suitability, Passives, Synergies & Resonance Sets!

- **Locations**: `js/martial.js:299, 303, 315, 329, 494, 536` vs `docs/ALL_EQUIPMENT_TIER_ART_JS2_V2_5_0.md` (`UI.resolveItemArt`)
- **Root Cause**:
  - Across the `898`-item equipment catalog, expansion packs authored items with raw `gearType` / `subType` tags such as **`'HEAVY_ARMOR'`**, **`'LIGHT_ARMOR'`**, and **`'HEADGEAR'`** (alongside `'PENDANT'`, `'ORNAMENT'`, `'SHOES'`, `'AUX_TOOL'`, `'SABER'`, `'GAUNTLET'`).
  - In `JS2-v2.5.0`, `UI.resolveItemArt` added a `typeAliases` table **solely for icon/aura rendering**, explicitly leaving gameplay untouched.
  - Meanwhile, **`M.gearType(it)` (`js/martial.js:299`)** still returns raw `it.gearType || it.weaponType || it.armorType || it.subType` **with zero alias normalization**!
- **Five Cascading Gameplay Failures Whenever an Armor Has `'HEAVY_ARMOR'`, `'LIGHT_ARMOR'`, or `'HEADGEAR'`**:
  1. **`M.gearSuitability` (`martial.js:536`)**: `armor.includes(type)` checks `['HEAVY']`/`['LIGHT']`/`['ROBE']` $\to$ evaluates to **`false`** (**`fit = 0.85`** `"เกราะต่างแนว"` **instead of `1.25` — a `-32%` suitability penalty on matching armor!**).
  2. **`M.evaluateSubEquipmentBonus` (`martial.js:494`)**: `prefA.includes(aType)` $\to$ **`false`** (**`0%` subpath armor bonus!**).
  3. **`M.weaponConditionMatches` (`martial.js:303`)**: Any passive/art requiring `armorTypes: ['HEAVY']` or `['LIGHT']` **deactivates**!
  4. **`M.evaluateSynergies` (`martial.js:315`)**: Subpath equipment synergy **fails**!

5. **`M.itemMatchesResonanceSet` (`martial.js:329`)**: Cannot match `rs.slots.ARMOR` in any of the 62 Resonance Sets unless `resonanceSetId` is hardcoded!

- **Exact Fix**: Add a canonical `M.TYPE_ALIASES` map right inside `M.gearType(it)` (`js/martial.js:299`):

```javascript
const TYPE_ALIASES = {
  HEAVY_ARMOR: 'HEAVY', LIGHT_ARMOR: 'LIGHT', HEADGEAR: 'HEAVY',
  SABER: 'BLADE', GAUNTLET: 'FIST', FIST_PALM: 'FIST',
  PENDANT: 'NECKLACE', ORNAMENT: 'CHARM', SHOES: 'CHARM', AUX_TOOL: 'CHARM'
};
```

(while preserving `it.visualSubtype` so `UI.resolveItemArt` still renders distinct Helmet/Boot/Pendant art!).

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2. `[ARM-LEG-02]` Quadruple Live Bug Across All 3 Legendary Armors (`visitors_clues.js:490–596, 730–940`)

Inspecting `VC.runWeaponOnStruckProcs` (`js/visitors_clues.js:918–940`) and `VC.runWeaponProcs` (`lines 725–908`) proves that **all three Tier-7 Legendary Armors execute copy-pasted Jade Dog-Beating Staff logic instead of their authored effects**:

1. **Wrong Combat Log Name on All 3 Armors (`line 933`)**:
   - Every defensive proc calls `COMBAT.applyDamage(W, f, defender, attacker, redirDmg, 'พลองหยกยืมแรงปัดสวน')`—printing *"Jade Dog-Beating Staff parry"* when the wearer is actually wearing a **Celestial Crane Robe**, **Shadow Kirin Cuirass**, or **Black Tortoise Heavy Plate**!
2. **`LEG_ARM_CELESTIAL_CRANE_ROBE` (อาภรณ์ขนนกกระเรียนสวรรค์เก้าเมฆา — `ROBE`)**:
   - **Authored Spec (`lines 508–509`)**: `shieldHpPct: 0.18`, **`qiRestorePct: 0.05` (ฟื้นฟูปราณ *5%*)**, `grantBuff: 'swiftWind'` — **zero reflect/damage in spec!**
   - **Live Bug**: `runWeaponOnStruckProcs` **ignores `qiRestorePct: 0.05` (`0` Qi restored!)** and instead deals phantom `15%` Dog-Beating Staff reflect damage + `'slow'`! Also never triggers *"หรือร่ายวิชา"* (`isSkill` in `runWeaponProcs`).
3. **`LEG_ARM_IMMORTAL_TORTOISE_PLATE` (เกราะหนักกระดองเต่าดำเสวียนอู่ค้ำสวรรค์ — `HEAVY`)**:
   - **Authored Spec (`lines 581–582`)**: `shieldHpPct: 0.24`, ล้างสถานะเกราะแตก/มึนงง, `grantBuff: 'aegisWard'`, `reflectRatio: 0.18`, `targetStatus: 'slow'`.

- **Live Bugs**:
    - **Never cleanses `'armorbreak'` or `'stun'` (`0` `ST.clearStatus` calls!)**.
    - Falls through to `ST.addStatus(defender, p.selfStatus || 'evadeUp', 7)` (`line 936`), granting **`'evadeUp'` (nimble evasion!)** to an immovable Black Tortoise Heavy Plate colossus!
4. **`LEG_ARM_SHADOW_KIRIN_CUIRASS` (เกราะอ่อนเกล็ดกิเลนเงาดารา — `LIGHT`)**:
    - **Authored Spec (`lines 544–545`)**: *"เมื่อถูกโจมตีหรือออกท่าสังหาร"* with `bonusDmgMult: 0.15, reflectRatio: 0.16, targetStatus: 'bleed', grantBuff: 'fervorMight'`.
    - **Live Bug**: `VC.runWeaponProcs` has **no branch** for `EFX_LEG_KIRIN_PHANTOM_COUNTER`, so `bonusDmgMult: 0.15` (`+15%` offensive assassination burst) never triggers!
5. **Stale `preferredSubs` & Missing 31 Grand Dao Paths (`lines 503, 540, 576`)**:
    - Replace stale IDs (`GRAND_FORMATION_ARCHITECT`, `SPIRIT_THREAD_PUPPET`, `SYMBIOTIC_FANG_FUSION`, `IRON_COLOSSUS`, `IRON_LEGION_MARSHAL`) with live IDs (`EARTH_VEIN_WARDEN`, `PHANTOM_SILK_PUPPETEER`, `SYMBIOTIC_BEAST_WARRIOR`, `HEAVENLY_GOLEM`, `BATTLEFIELD_STRATEGIST`) and add Grand Dao tanks/assassins/casters (`UNBROKEN_IRON_STANCE`, `BLACK_TORTOISE_SHELL_COLOSSUS`, `SMOKE_MIST_ASSASSIN`, `GOLDEN_LOTUS_CHAN`, `STAR_CHART_PROPHET`)!

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3. `[ARM-ADAPT-03]` Universal Permanent Adaptive-Armor Leak (`ECO.adaptArmorToElement`, `js/economy.js:1100–1111`)

- **Exact Code (`economy.js:1100–1111`)**:

```javascript
arm.adaptiveState = arm.adaptiveState || { lastElement: null, stacks: 0,
resistBonusPct: 0 };
if (arm.adaptiveState.lastElement === el) arm.adaptiveState.stacks = Math.min(3,
arm.adaptiveState.stacks + 1);
...
arm.adaptiveState.resistBonusPct = +(arm.adaptiveState.stacks * 0.10).toFixed(2);
```

- **Two Logical & Balance Bugs**:
    1. **Every Tier-1 Rag Has Mythic Elemental Adaptation**: `COMBAT.applyDamage` calls `adaptArmorToElement` on **every** `d.equip.ARMOR` without checking

whether the armor actually has an Adaptive trait/effect or `tier >= 5`! Even a 10-stone Tier-1 cloth tunic gains **`-30%` elemental damage taken** after 3 hits!

2. **Infinite Persistence Across Years & Battles**: `arm.adaptiveState` is mutated directly onto the permanent item object and **never cleared in `COMBAT.leave` (`combat.js`)** (and even serialized into save files)! Getting hit 3 times by a weak Fire Sprite pre-charges `-30%` Fire Resistance before Round 1 of a future Boss fight!

- **Fix**: Store transient combat adaptation on the combatant (`d._combatArmorAdapt = { el, stacks, until: W.t + 18 }`, cleared in `COMBAT.leave`), cap ordinary T1–T4 armors at `0%` (reserving `5%..10%/stack` up to `3` stacks specifically for armors with `EFX_ADAPTIVE_WEAVE` / `ELEMENTAL_ADAPT` or `tier >= 6`)!

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4. `[ARM-ARTISAN-04]` `ECO.applyArmorArtisan` Bonus Wiped Clean by Rune Socketing or Stat Regeneration (`js/economy.js:216–226` vs `1294`)!

- **Exact Code Bug**:
  - `ECO.applyArmorArtisan` (`economy.js:219–220`) multiplies `gear.st.def` and `gear.st.mdef` by `(1 + bonus)` and sets `gear.artisan = { ... }`, **without updating `gear.baseSt` (`economy.js:702`)**!
  - Later, if a smith engraves Elemental Runes (`ECO.socketElementalRunes`, `economy.js:1294`) or evolves gear (`delete gear.st; ST.gearStats(gear)`), `ST.gearStats` rebuilds `gear.st` from `gear.baseSt` (or base tier formula), **silently erasing the `CELESTIAL_ARMOR_ARTISAN` `+8%` defense bonus** while `gear.artisan` blocks re-application!
- **Fix**: Also multiply `gear.baseSt.def` and `gear.baseSt.mdef` when `gear.baseSt` exists, **and** have `ST.gearStats(gear)` apply `gear.artisan.defenseBonus` deterministically whenever regenerating `gear.st`!

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5. `[ARM-ARCH-05]` Authored Armor Archetype Identity, Mutual Damage-Share Loop Risk & Asymmetric Tank Wear

1. **Authored Armors (`ITEM190_EQ_009..013`, `REC140_FRG_PELT_ARMOR_2`, `REC140_FRG_IMMORTAL_ARMOR_6` in `data_items.js`) Missing `gearType` & `move`/`cast`/`eva` Tradeoffs**:

- Explicitly stamp `gearType`: `ITEM190_EQ_009` (เกราะอ่อนไหมฟ้า) $\to$ `'LIGHT'`, `ITEM190_EQ_010` (เกราะสำริดหนาม) $\to$ `'HEAVY'`, `ITEM190_EQ_011` (เสื้อคลุมขนพยัคฆ์) $\to$ `'ROBE'`, `ITEM190_EQ_012` (เกราะเกล็ดเต่ามังกร) $\to$ `'HEAVY'`, `ITEM190_EQ_013` (เกราะทองคำวัชระ) $\to$ `'HEAVY'`.
- Ensure `ST.gearStats` gives authored `HEAVY` (`+def, -move`), `LIGHT` (`+eva, +move`), and `ROBE` (`+mdef, +cast`) the same tactile weight/agility feel as procedural armors!

2. **Recursion & Range Guard on `ITEM190_EQ_012` (`EFX_GUARDED_DAMAGE_SHARE`) & `ITEM190_EQ_013` (`EFX_REALM_SUPPRESSION_GUARD`)**:
   - If two frontline tanks both equip `ITEM190_EQ_012` (`shareRatio: 0.35`), damage-sharing MUST pass `{ noShare: true, noReflect: true }` and require `unitDist <= 3.5` so two tanks never bounce shared damage back and forth or absorb hits from 40 tiles away!
   - On `ITEM190_EQ_013`, compute `crossRealmDelta` using **`CULT.normalizedStage(attacker) - CULT.normalizedStage(defender)`** (never raw `stage`, which breaks against 5-stage `BODY` / 4-stage `GOD` bosses!).
3. **Asymmetric Frontline Tank Armor Wear & Non-Procedural `dur = 1` (`30%` Stat Rot)**:
   - A solo `FRONT_ROW` tank takes `4–6×` more hits per battle than backline mages (`-0.24 dur` per hit), burning through procedural `HEAVY` armor durability rapidly while authored/Legendary armors decay in `SIM.hourly` to `dur = 1` (`df = 0.30`) with zero NPC auto-repair!
   - Scale `HEAVY` armor base durability resilience (`0.16` wear/hit vs `0.24` on `ROBE`), exempt `legendary` armors from permanent `maxDur` shrinkage, and teach `ECO.pickOrder` (`economy.js:449`) to count **broken/low-`maxDur` frontline armors** as `gapA` demand so the Forge proactively crafts replacement armor for sect guards!

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6. Implementation & Verification Checklist for GPT / Gemini (`05A`)

- [ ] **Normalize Raw Equipment Tags in `M.gearType` (`js/martial.js:299`)**: Map `HEAVY_ARMOR -> HEAVY`, `LIGHT_ARMOR -> LIGHT`, `HEADGEAR -> HEAVY`, `PENDANT -> NECKLACE`, `ORNAMENT/SHOES/AUX_TOOL -> CHARM`, `SABER -> BLADE`, `GAUNTLET/FIST_PALM -> FIST`.
- [ ] **Fix All 3 Legendary Armors (`js/visitors_clues.js:730–940`)**: Separate `CRANE_ROBE` (restore `+5%` Qi, `0` staff reflect, own log text), `TORTOISE_PLATE` (cleanse `armorbreak`+`stun`, no `evadeUp`, own log text), and `KIRIN_CUIRASS` (wire offensive `+15%` `bonusDmgMult` in `runWeaponProcs`, own log text); update `preferredSubs`.

- [ ] **Scope & Reset Adaptive Armor (`economy.js:1100`, `combat.js`)**: Reset adaptation on `COMBAT.leave` and restrict to Adaptive-tagged / T6+ armors.
- [ ] **Preserve `applyArmorArtisan` Across Rebuilds (`economy.js:216`, `stats.js`)**: Persist bonus in `baseSt` / `ST.gearStats`.
- [ ] **Stamp `gearType` on `data_items.js` Armors & Fix Tank Durability Loop**: Guard `EFX_GUARDED_DAMAGE_SHARE` recursion/range, use `CULT.normalizedStage` on `EFX_REALM_SUPPRESSION_GUARD`, and include worn-out armor in `ECO.pickOrder` forge demand.